

졸업프로젝트 면담

2026.04.29.(수)

Baseline Results

No.	Build Type	Input Dataset (#vertex)	Output	Runtime
1-1	Augmented Surface Interpolation	Humanoid_rm (2,000)	[level 0] init.ply 9개, curve.ply 9개	2시간 7분
1-2		Humanoid_new (2,818)	[level 0] init.ply 9개, curve.ply 9개	8분
2		Sphere Through Tunnel (2,562)	[level 0] init.ply 5개, curve.ply 5개 [level 1] init.ply 9개, curve.ply 9개	96분 (= lev_1 50분 + lev_2 46분)
3	Augmented Surface Extrapolation	Leaf (5,257)	comb_exp__ 0 .ply ~ comb_exp_122.ply	[조기 중단] 122/450 steps : 2시간 17분
4		Head Punch (4,354)	comb_exp_0.ply ~ comb_exp_32.ply	37분
5	Elastic Surface Extrapolation	Leaf (5,257)	comb_exp_0.ply ~ comb_exp_450.ply	1시간 15분
6		Head Punch (4,354)	comb_exp_0.ply ~ comb_exp_32.ply	3분
7	Repulsive Means	Mean_X (2,000)	init.ply, input.ply 3개, mean.ply 6개, tpe_mean.ply 6개	1시간 38분
8	Shape From Metric	Hyperbolic Disk (2,791)	에러 나서 실행 실패	
9		Torus (16,384)	undeformed.ply, deformed.ply	1시간 27분

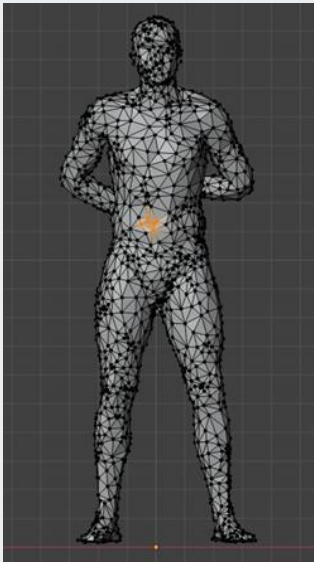
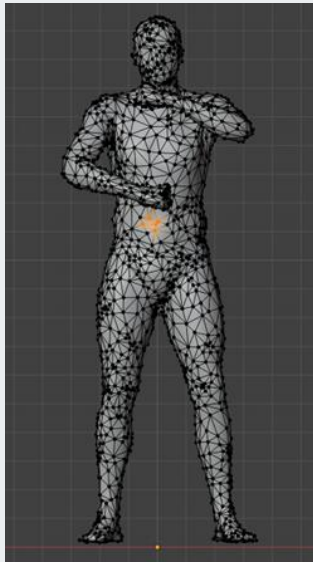
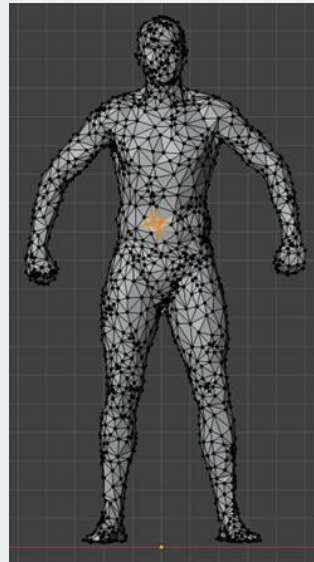
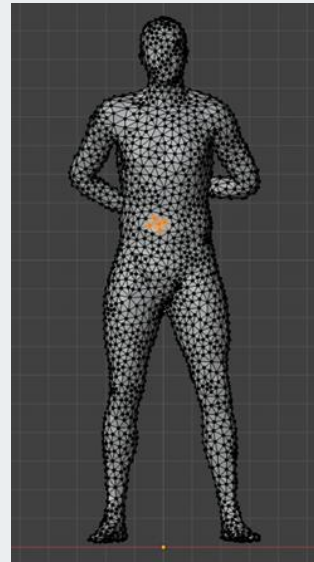
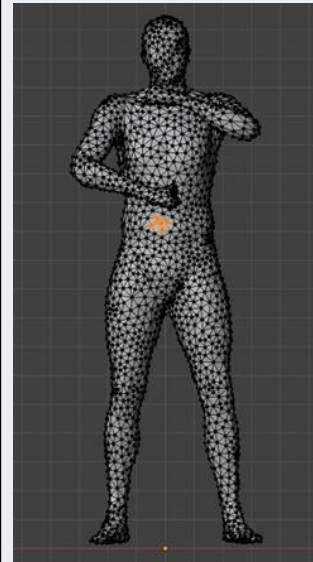
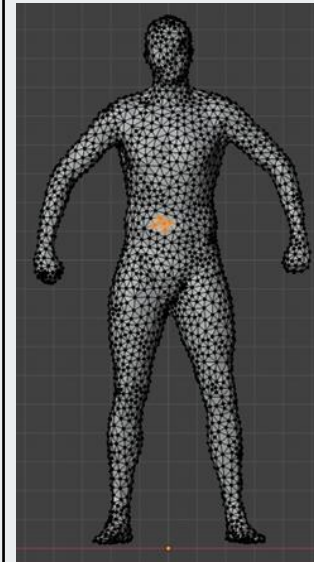
Baseline Input Data

- Augmented Surface Interpolation(1/2) - Humanoid

						
Input Type	start	end	init	start	end	init
Data Name	Rigging_SCAPE_rm_5.ply	Rigging_SCAPE_rm_6.ply	Rigging_SCAPE_rm_ref.ply	new5.ply	new6n.ply	new.ply
#Vertices	2,000			2,818		
#Edges	5,994			8,448		
#Faces	3,995			5,632		

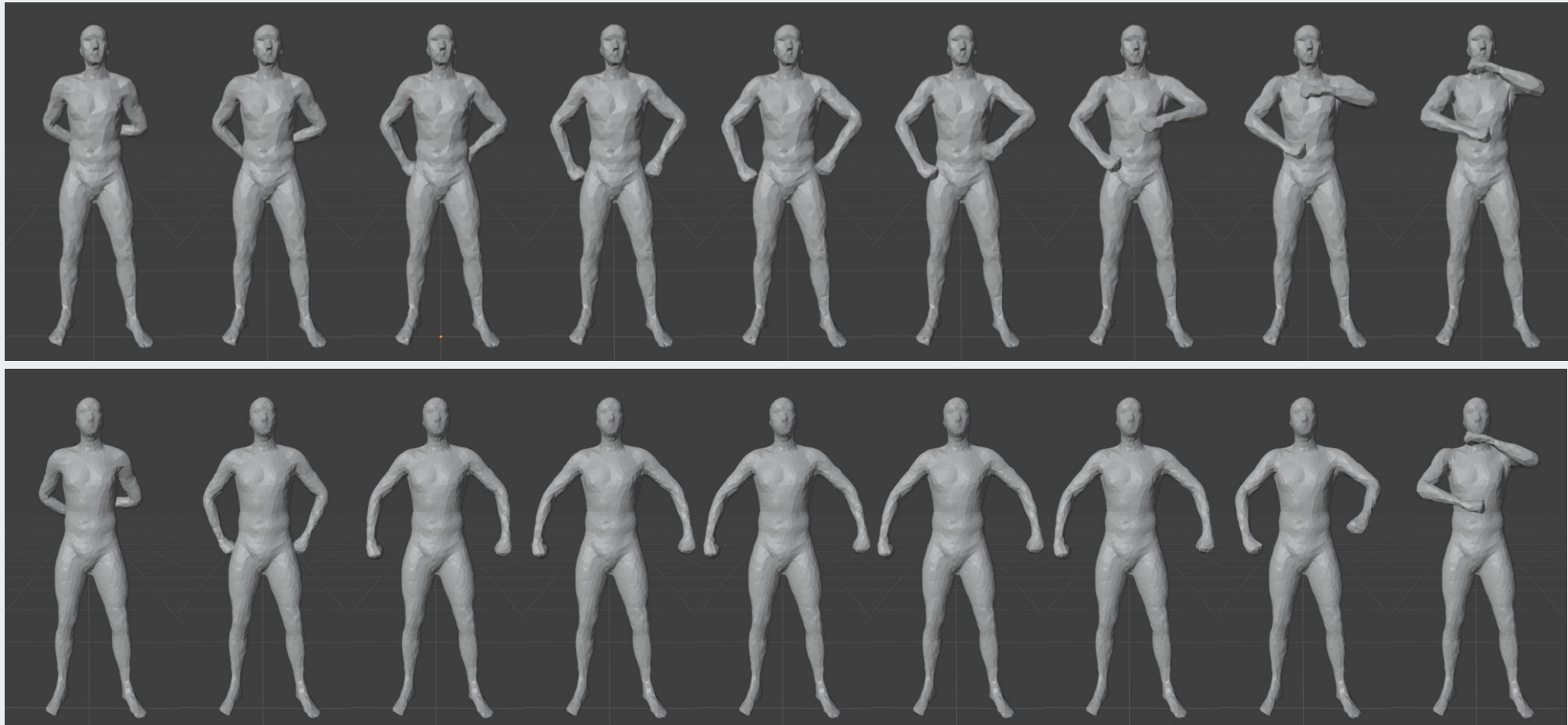
Baseline Input Data with Dirichlet Constraints (5 vertices)

- Augmented Surface Interpolation(1/2) – Humanoid

						
Input Type	start	end	init	start	end	init
Data Name	Rigging_SCAPE_rm_5.ply	Rigging_SCAPE_rm_6.ply	Rigging_SCAPE_rm_ref.ply	new5.ply	new6n.ply	new.ply
#Vertices	2,000			2,818		
#Edges	5,994			8,448		
#Faces	3,995			5,632		

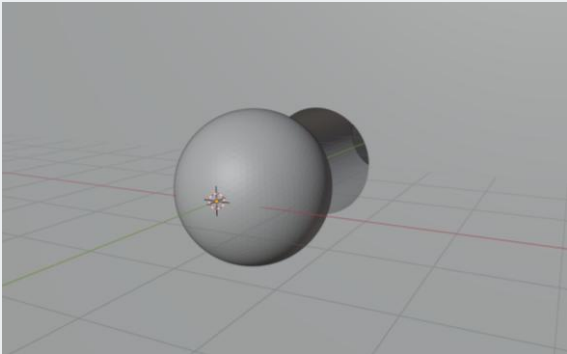
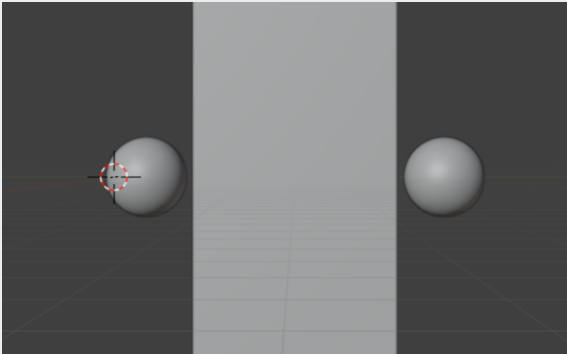
Baseline Output Data

- Augmented Surface Interpolation(1/2) – Humanoid
 - (위) #v=2,000 / Runtime 2시간 7분
 - (아래) #v=2,818 / Runtime 8분



Baseline Input Data

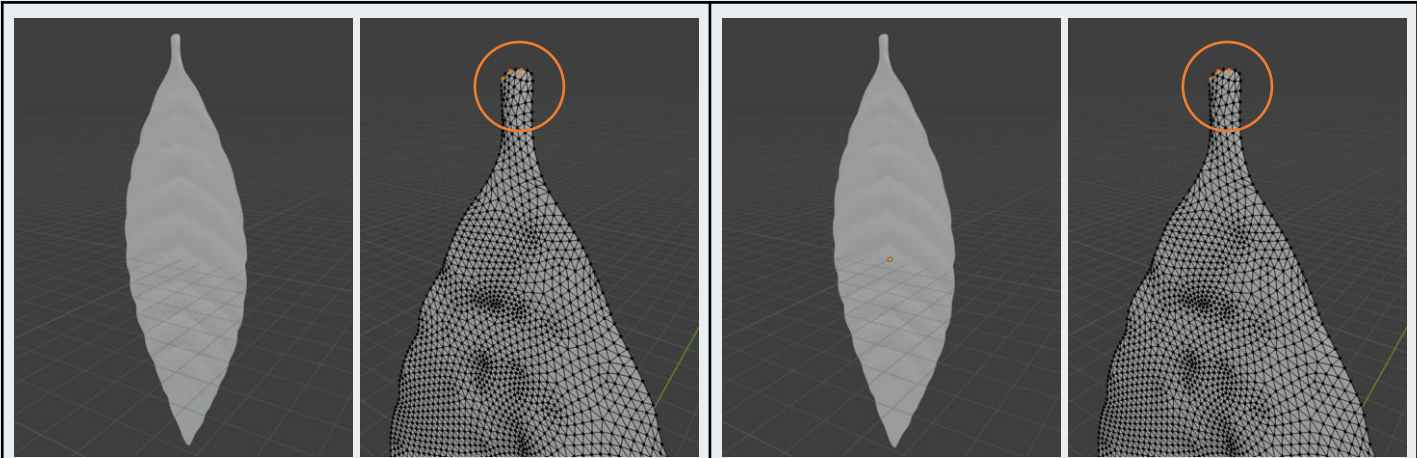
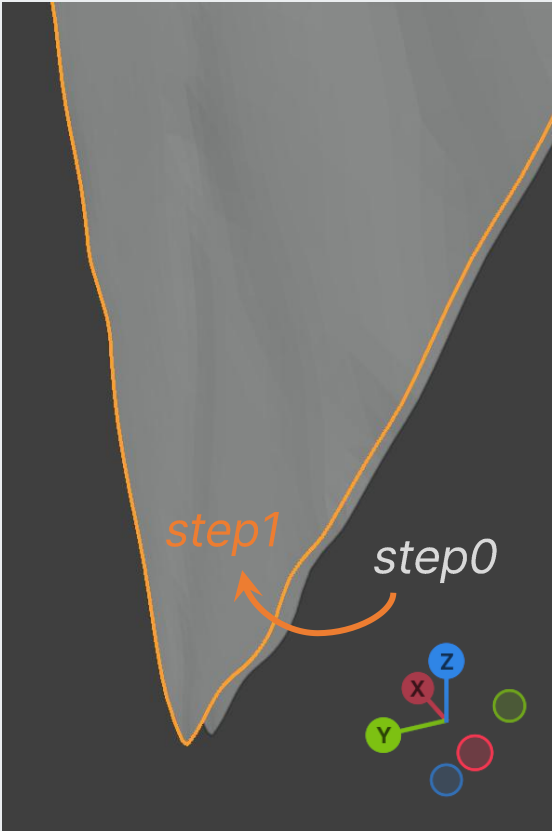
- Augmented Surface Interpolation(2/2) – Sphere Through Tunnel



	Input Type	Data Name	#Vertices	#Edges	#Faces
	start, init	TranslationLoop4Sphere_L1.ply	2,562	7,680	5,120
	end	TranslationLoop4Sphere_L2.ply			
	obstacle	Wall_7.ply	7,248	21,744	14,496

Baseline Input Data with Dirichlet Constraints (3 vertices)

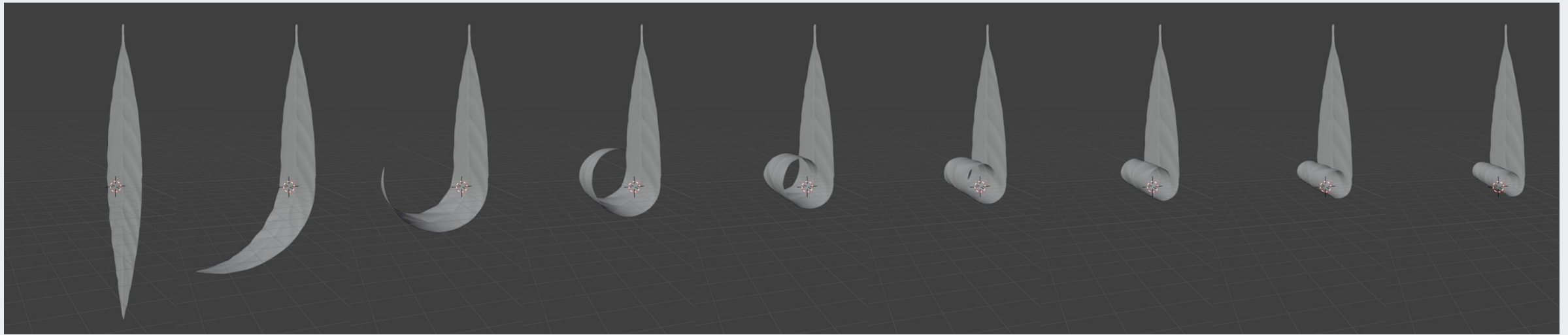
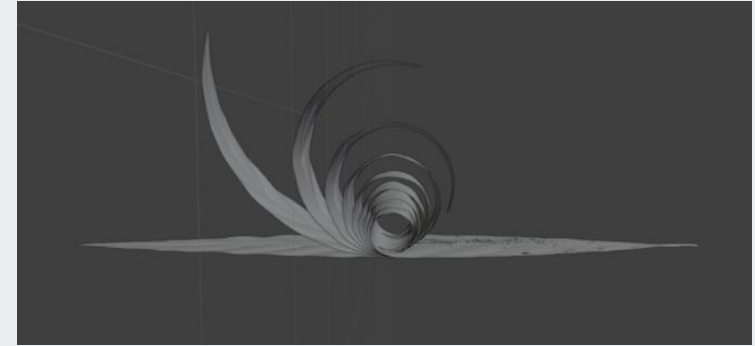
- Augmented Surface Extrapolation(1/2) – Leaf



Input Type	start	second
Data Name	sbdv_step0.ply	sbdv_step1.ply
#Vertices	5,257	
#Edges	15,448	
#Faces	10,192	

Baseline Output Data

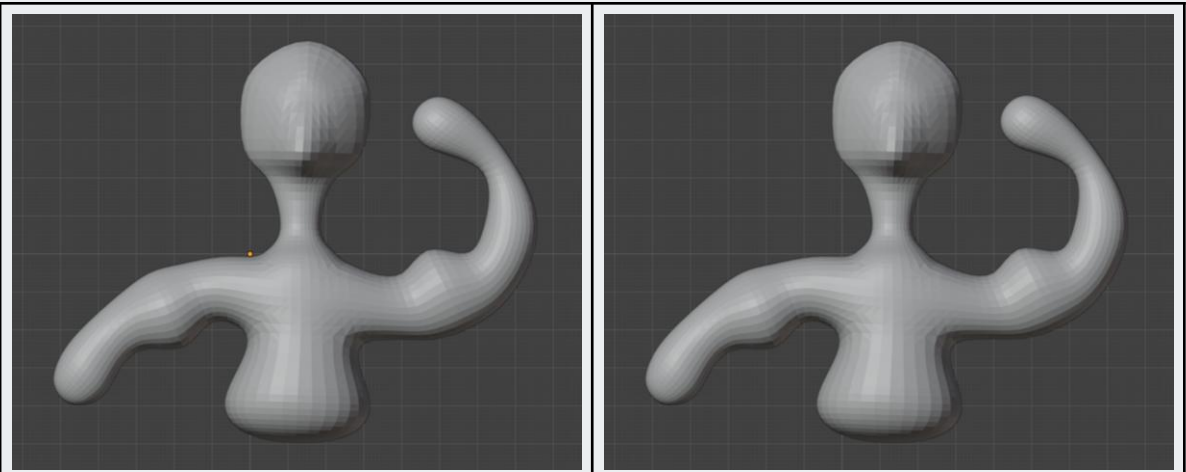
- Augmented Surface Extrapolation(1/2) – Leaf



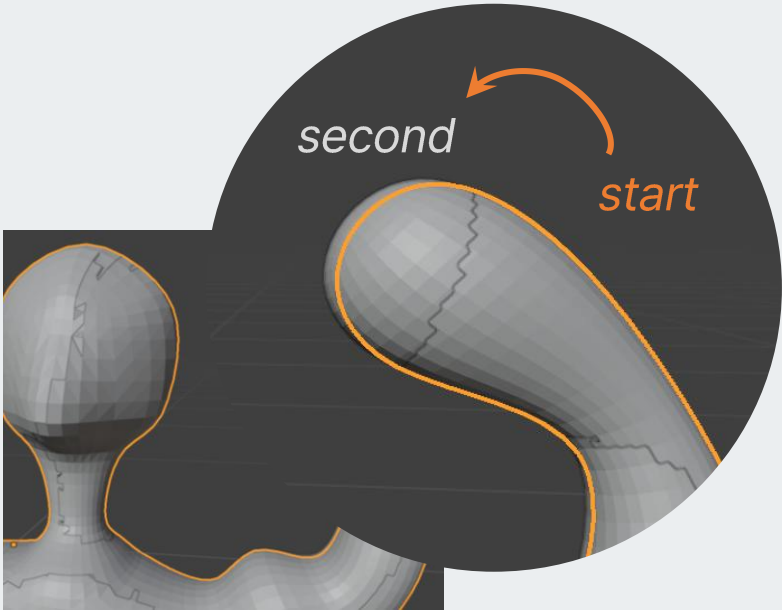
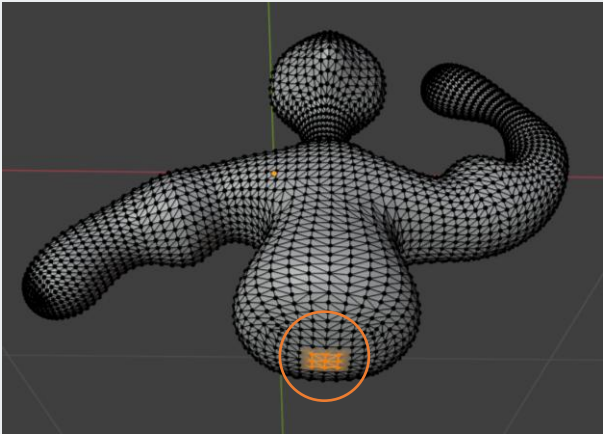
0 50 100 150 200 250 300 350 400

Baseline Input Data with Dirichlet Constraints (9 vertices)

- Augmented Surface Extrapolation(2/2) – Head Punch

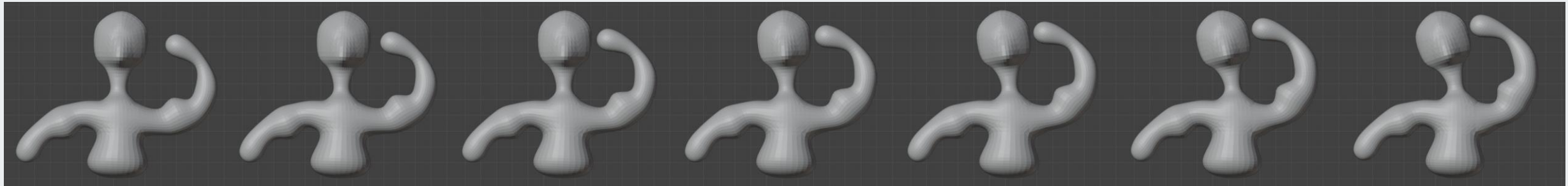
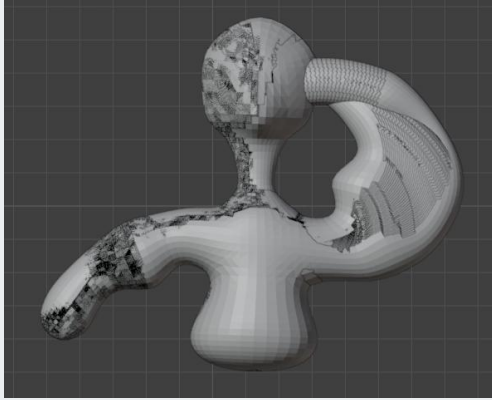


Input Type	start	second
Data Name	punch1.obj	punch3.obj
#Vertices	4,354	
#Edges	13,056	
#Faces	8,704	



Baseline Output Data

- Augmented Surface Extrapolation(2/2) – Head Punch



0

5

10

15

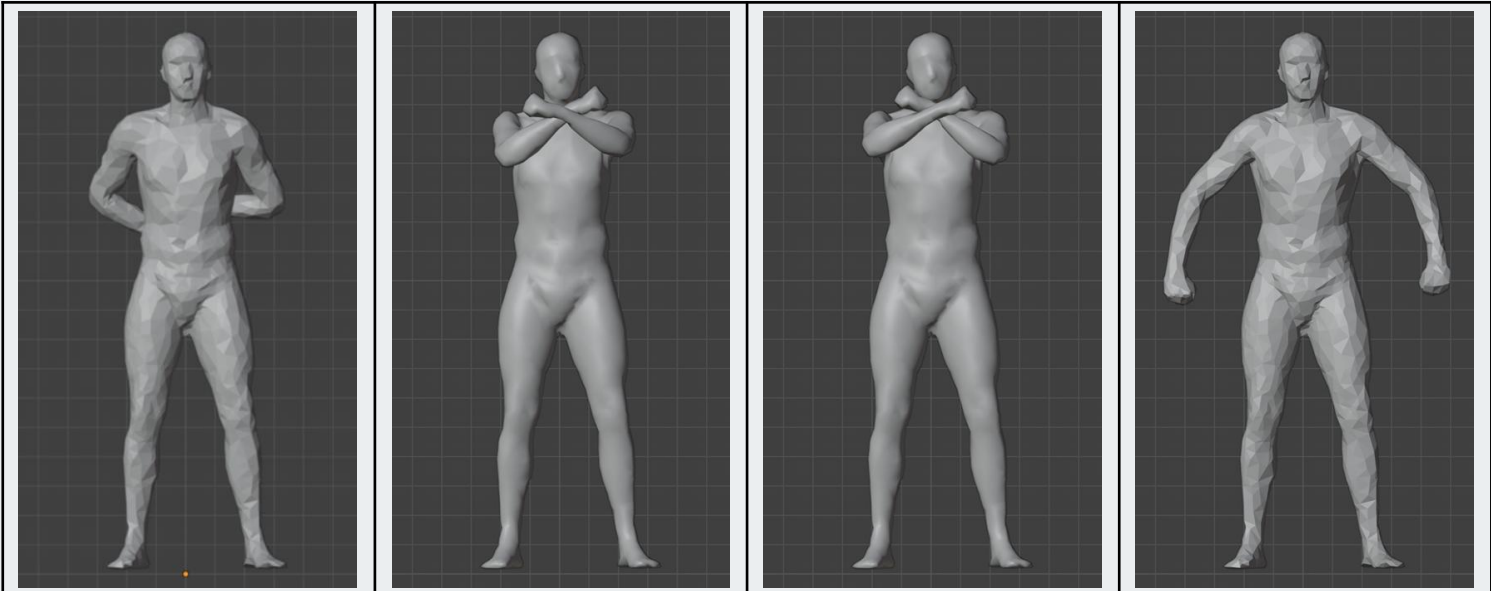
20

25

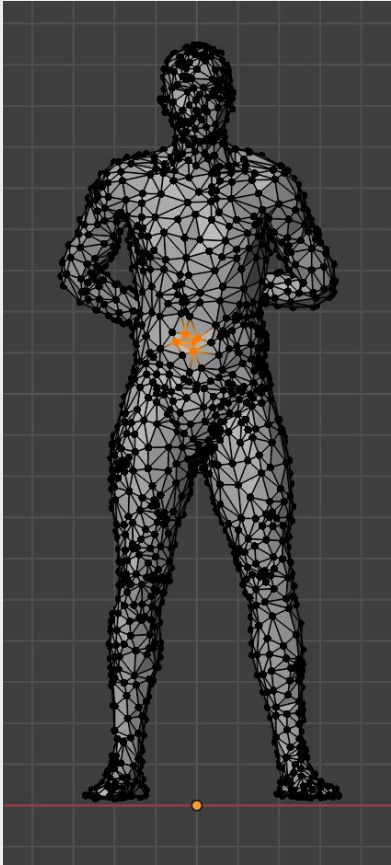
30

Baseline Input Data with Dirichlet Constraints (5 vertices)

- Repulsive Means – Mean_X

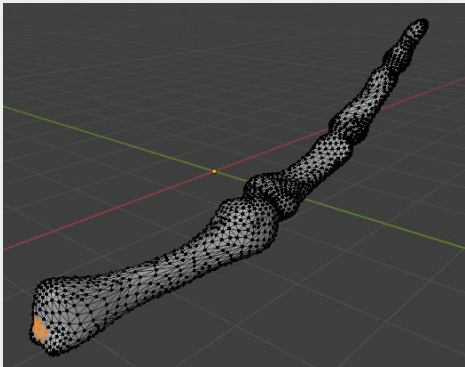
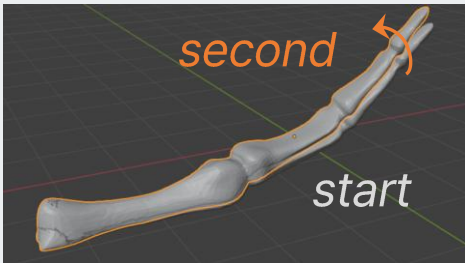
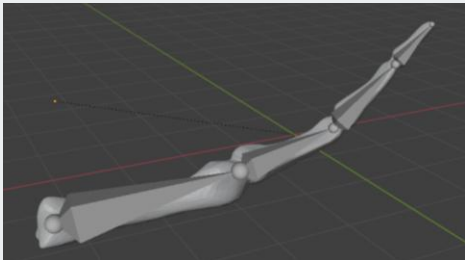


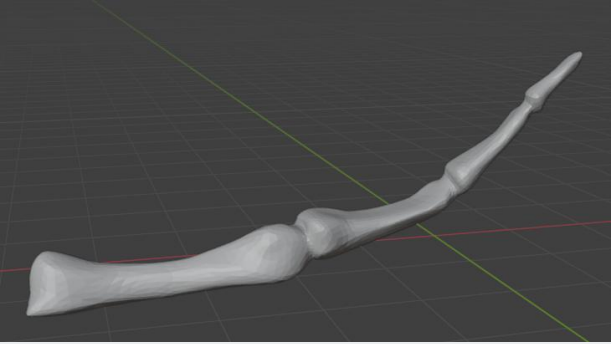
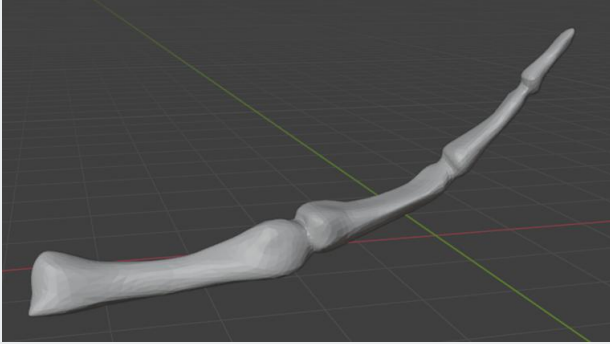
Input Type	input	input	input	init
Data Name	Rigging_SCAPE_rm_5	Rigging_SCAPE_rm_8	Rigging_SCAPE_rm_9	Rigging_SCAPE_rm_ref
#Vertices	2,000			
#Edges	5,994			
#Faces	3,995			



My Input Data with Dirichlet Constraints (7 vertices)

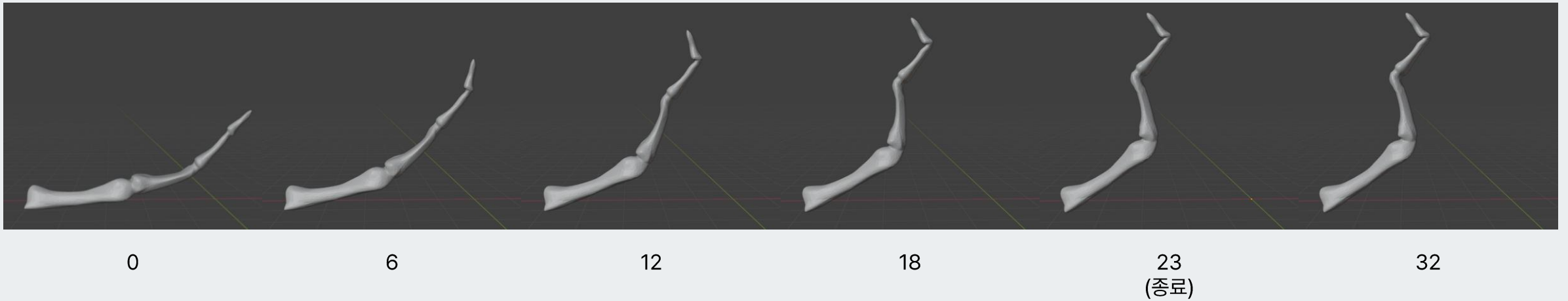
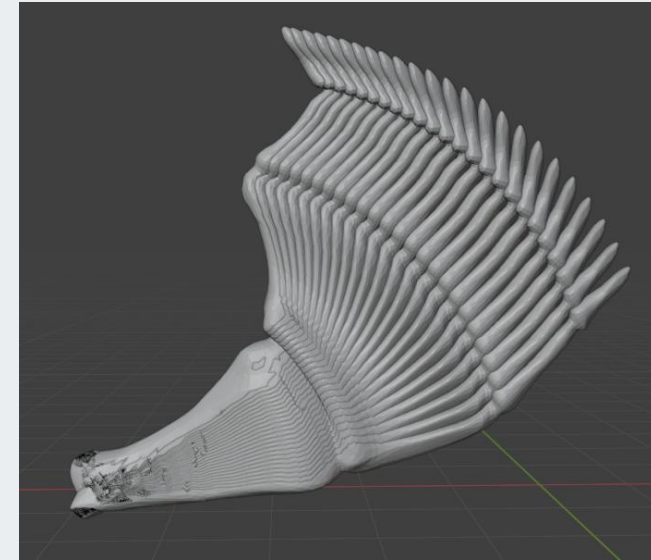
- Augmented Surface Extrapolation – Finger



		
Input Type	start	second
Data Name	finger_rigging_start.ply	finger_rigging_second.ply
#Vertices	2,046	
#Edges	6,132	
#Faces	4,088	

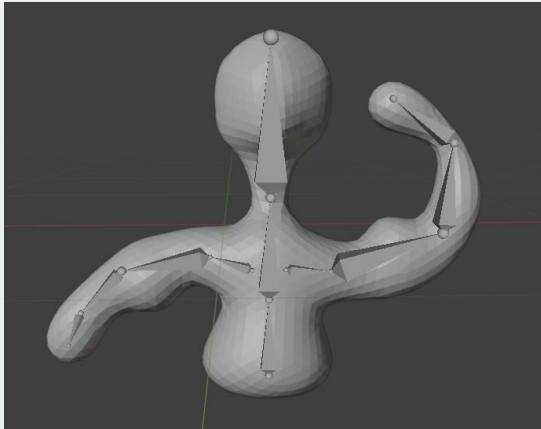
My Output Data

- Augmented Surface Extrapolation – Finger
 - numSteps 32로 설정
 - 24 step부터는 더 진행되지 않음
 - Self-Rigging 데이터라서 매끄럽지 않은 결과물이 나옴
 - Runtime 27분



My Input & Output Data

- Augmented Surface Extrapolation – Head Punch
 - 원본 init 데이터의 vertex 4,354개 → vertex 개수를 2,070개로 줄인 new_init 데이터
 - new_init 데이터를 Rigging 하여 new_second 데이터를 생성
 - Augmented Surface Extrapolation 진행



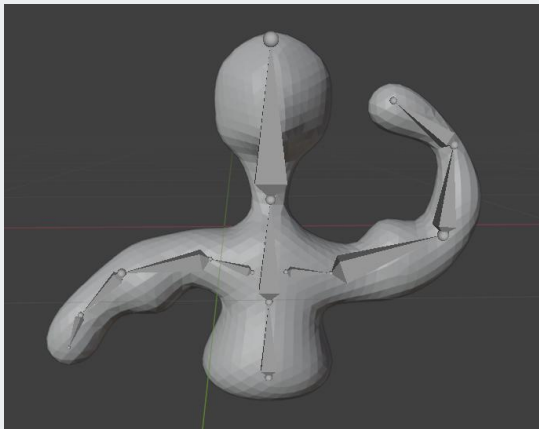
[Input] Rigging

My Input & Output Data

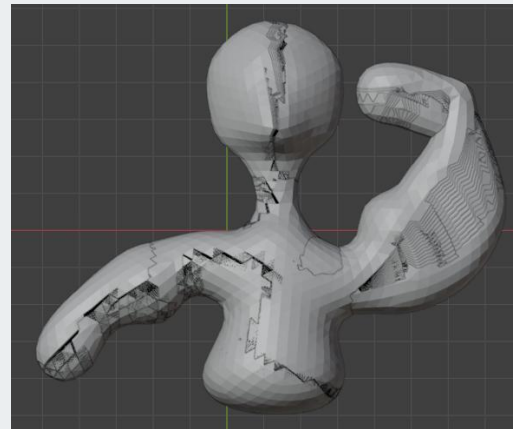
- Augmented Surface Extrapolation – Head Punch
 - Step 17부터는 수렴하지 않음 (vs. 원본 데이터는 Step 32까지 진행됨)
 - Self-Rigging 데이터라서 매끄럽지 않은 결과물이 나옴
 - Runtime 28분 (Step 0~16)
- * Blender에서 raw 데이터 비교

```
step = 22, stepsize = 5.214465e-01, norm = 1.882542e-08, t = 3511.423910ms
step = 23, stepsize = 5.415303e-01, norm = 1.208413e-08, t = 1288.052476ms
step = 24, stepsize = 2.034049e-03, norm = 1.206805e-08, t = 1654.122359ms
step = 25, stepsize = 0.000000e+00, norm = 1.206805e-08, t = 2892.333308ms
=====
Finished Newton's method after 25 steps (max. steps = 10000, tol = 1.000000e-08).
Final stepsize = 0.000000e+00, final norm = 1.206805e-08
=====
```

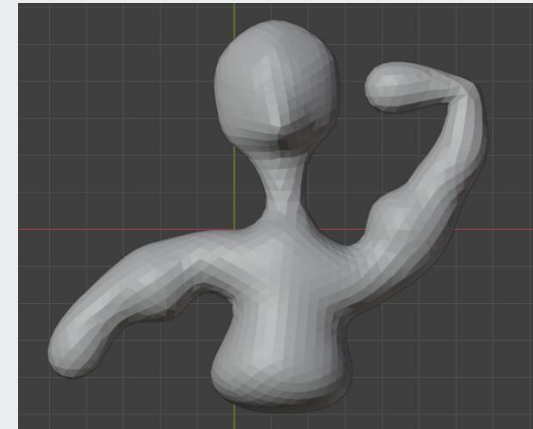
```
step = 713, stepsize = 3.963391e-06, norm = 1.060768e-07, t = 2901.286183ms
step = 714, stepsize = 3.963395e-06, norm = 1.060765e-07, t = 2206.386908ms
step = 715, stepsize = 3.963391e-06, norm = 1.060763e-07, t = 1942.249470ms
step = 716, stepsize = 3.963390e-06, norm = 1.060760e-07, t = 2198.137144ms
step = 717, stepsize = 3.963393e-06, norm = 1.060758e-07, t = 2027.749822ms
step = 718, stepsize = 3.963392e-06, norm = 1.060755e-07, t = 2201.998386ms
step = 719, stepsize = 3.963388e-06, norm = 1.060753e-07, t = 2127.338884ms
step = 720, stepsize = 3.963394e-06, norm = 1.060750e-07, t = 2053.960420ms
```



[Input] Rigging



[Output] 0~16



[Output] 16

Problematic Cases

- Augmented Surface Extrapolation

- 1) Leaf 데이터 Rigging 문제 (* Blender 확인)

- Augmented Surface Interpolation

- 1) Baseline의 Runtime이 차이나는 이유 (2시간 7분 vs. 8분)
 - 2) Humanoid 데이터의 vertex 개수를 1,048개로 줄인 버전
 - 3) dirichletWeight 값을 1로 설정한 버전
 - 4) 그 외 각각의 hyper-parameter 효과를 확인하기 위해 하나씩 변경하기 등
- 수렴이 되지 않아서 계속 진행하거나 오류가 발생해서 조기 종료됨
 - Extrapolation과 다르게, 종료 직전에 한꺼번에 결과 파일이 저장되기 때문에 중간 과정을 확인하기 어려움. (코드 수정?)
 - [질문] hyper-parameter의 효과를 확인하기 위해서는 튜닝하여 결과를 확인해보는 방법이 유일한지?
 - [질문] 순차적으로 변하는 mesh 데이터를 ppt에 효과적으로 시각화 하는 방법 (사진, 영상 등)